



# Abductive multi-instance multi-label learning for periodontal disease classification with prior domain knowledge

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## ABSTRACT

Machine learning is widely used in dentistry nowadays, offering efficient solutions for diagnosing dental diseases, such as periodontitis and gingivitis. Most existing methods for diagnosing periodontal diseases follow a two-stage process. Initially, they detect and classify potential Regions of Interest (ROIs) and subsequently determine the labels of the whole images. However, unlike the recognition of natural images, the diagnosis of periodontal diseases relies significantly on pinpointing specific affected regions, which requires professional expertise that is not fully captured by existing models. To bridge this gap, we propose a novel **AB**ductive **M**ulti-**I**nstance **M**ulti-**L**abel learning (**AB-MIML**) approach. In our approach, we treat entire intraoral images as “bags” and local patches as “instances”. By improving current multi-instance multi-label methods, AB-MIML seeks to establish a comprehensive many-to-many relationship to model the intricate correspondence among images, patches, and corresponding labels. Moreover, to harness the power of prior domain knowledge, AB-MIML converts the expertise of doctors and the structural information of images into a knowledge base and performs abductive reasoning to assist the classification and diagnosis process. Experiments unequivocally confirm the superior performance of our proposed method in diagnosing periodontal diseases compared to state-of-the-art approaches across various metrics. Moreover, our method proves invaluable in identifying critical areas correlated with the diagnosis process, aligning closely with determinations made by human doctors.

## 1. Introduction

Dental diseases pose a significant challenge to oral health, affecting a substantial portion of the global population. According to the WHO Global Oral Health Status Report (WHO, 2022), it is estimated that oral diseases impact nearly 3.5 billion people worldwide. Among them, periodontitis is receiving tremendous attention, as it is reported to affect approximately 40% of individuals over the age of 30 (Eke et al., 2016), and has been linked to various medical conditions, including diabetes mellitus (Winning et al., 2017). Therefore, the prevention and diagnosis of periodontitis have become prominent topics in modern medical research, which influence the health of billions of people. Based on distinct clinical manifestations, periodontal diseases can be classified into various types, such as gingivitis and periodontitis. Traditional diagnosis of periodontal diseases typically involves patient history and clinical examination, such as assessments of pocket depth

and evaluation of alveolar bone loss (Årtun and Urbye, 1988; Seabra et al., 2008; Ramachandra et al., 2009; Preshaw, 2015). However, these approaches are often invasive, and the measurements can vary among examiners depending on the probes used (Osborn et al., 1992; Hefti, 1997). Therefore, developing a novel method is crucial to help standardize the diagnostic process.

The advent of Artificial Intelligence (AI) in dentistry has revolutionized the diagnosis of periodontal diseases. Extensive research has demonstrated the potential of AI in several aspects of periodontal disease management (Muresan et al., 2020; Farook et al., 2023; Cholan et al., 2023). Other studies have also explored the application of machine learning in the periodontal diagnosis (Özden et al., 2015; Chen et al., 2018; Feres et al., 2018; Hung et al., 2019; Na et al., 2020; Farhadian et al., 2020). While these methods have shown great promise in early AI-assisted periodontal diagnosis, their effectiveness appears limited when addressing high-resolution images and real-world

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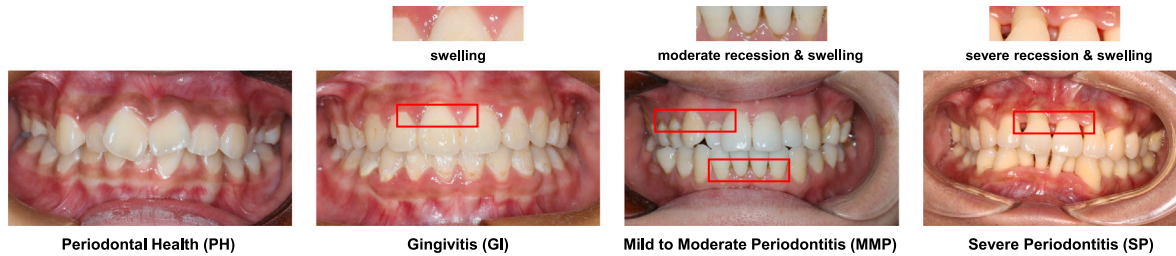


Fig. 1. Example images of four concerned categories of periodontal diseases. The red boxes highlight the typical affected patches, such as the swelling area of GI, the moderate recession & swelling area of MMP, and the severe recession & swelling area of SP. They are all located in some specific local patches of the original images.

medical scenarios. With the rise of deep learning, numerous model-based methods have been developed to create an end-to-end diagnostic loop. Intraoral color images or X-rays serve as essential data sources for accurate assessments through deep learning models. However, due to the inherent structure of human teeth, only a few regions may exhibit notable changes indicating inflammation or disease. Therefore, a two-stage process is widely employed based on effective object detection methods, such as RCNN (Girshick et al., 2014; Girshick, 2015; Ren et al., 2015) and YOLO (Redmon et al., 2016; Redmon and Farhadi, 2017, 2018; Bochkovskiy et al., 2020; Wang et al., 2023). This approach generally follows a specific framework: Initially, the detection model learns from the training data to extract and classify potentially affected patches; Subsequently, these patches (or their embeddings) are used as inputs to determine the labels of the whole images (Chang et al., 2020; Li et al., 2021; Ding et al., 2021; Öziç et al., 2023). Leveraging the powerful capabilities of deep learning models, these methods have achieved notable results in their tasks, with accuracy and recall rates comparable to those of human doctors.

Despite their impressive performance in classifying periodontal diseases, the aforementioned methods have certain limitations. In terms of disease modeling, there is a lack of research that integrates the types and severity of periodontal diseases for comprehensive analysis. Most studies treat all forms and degrees of periodontal diseases as parallel, overlooking potential hierarchical and ordinal correspondences in classification criteria. Consider the four main categories recognized by doctors: periodontal health (PH), gingivitis (GI), mild to moderate periodontitis (MMP), and severe periodontitis (SP). Some examples of the four categories are shown in Fig. 1. Through expert knowledge, we can safely reduce these labels to two distinct dimensions: the degree of gingival swelling and recession. The relationship between the original labels and the reduced dimensions can be described as follows: The gingivitis group exhibits gingival swelling without gingival recession; the mild to moderate periodontitis group displays both gingival swelling and mild to moderate gingival recession, while the severe periodontitis group shows severe gingival recession alongside gingival swelling. Hence, it is clear that these categories cannot be treated as completely independent. Instead, novel modeling approaches are needed to capture the intricate relationships between inputs and categories. Moreover, the diagnosis of periodontal disease often requires a comprehensive consideration of suspected areas and periodontal conditions of adjacent regions, which detection-based methods often fail to accomplish. Additionally, detection-based methods require a substantial amount of finely annotated data to accurately locate possible ROIs. Such fine-grained annotation requires doctors to carefully and precisely examine each image to identify all potentially affected areas without any omissions, which is hard and time-consuming. Furthermore, since the periodontal structure remains consistent across individuals, affected areas often manifest in specific regions, placing significant reliance on doctors' professional knowledge. Unfortunately, existing methods do not effectively incorporate this critical prior information into the training process.

To address these issues, in this paper, we propose a novel ABductive Multi-Instance Multi-Label Learning (AB-MIML) approach for the

classification and diagnosis of periodontal disease on coarse-grained annotated data. In this task, we mainly deal with four types of labels: periodontal health (PH), gingivitis (GI), mild to moderate periodontitis (MMP), and severe periodontitis (SP). To model the intricate relationships between inputs and results, we employ an improved multi-instance multi-label approach. Specifically, we treat an image as a “bag” containing multiple “instances” corresponding to smaller patches. By dividing the input image into these patches, we aggregate patch-level predictions using attention-based techniques and map them to the reduced labels to uncover fine-grained connections. Additionally, we integrate prior domain knowledge into the classification process. This relevant prior knowledge is expressed as logical formulas to create a knowledge base. Abductive reasoning is then utilized to derive the most probable outcomes, correcting classification errors and generating patch-level labels based on implied logical relationships. Due to the lack of publicly available datasets in this field, we collect and annotate a dataset from Nanjing Stomatological Hospital to validate the effectiveness of our proposed method. The results demonstrate that our method significantly enhances classification performance across various metrics compared to other baseline methods. Furthermore, we showcase our method's ability to identify crucial diseased patches through specific examples, thereby enhancing the model's interpretability. To conclude, our major contributions are:

- **We propose the AB-MIML approach for conducting fine-grained modeling for the classification of periodontal diseases.** Since specific patches significantly influence the predicted results, and disease labels can be reduced to finer dimensions, we transform the problem into a many-to-many machine learning task, and employ an improved multi-instance multi-label method to capture the complex correlations between input images and corresponding disease labels.
- **Our proposed AB-MIML approach successfully incorporates prior domain knowledge into the classification process in the field of medical imaging for the first time.** By abductive reasoning, we transfer prior domain knowledge into first-order formulas and generate most probable abduced labels to help correct classification errors and identify key patches. Consequently, AB-MIML improves accuracy at both image and patch levels with coarse-grained annotation.
- **Experimental results demonstrate the superiority of our AB-MIML approach.** We conduct experiments comparing various methods, and the results demonstrate that our approach significantly outperforms baseline models. Moreover, by identifying patches that have a greater impact on the classification results in a more accurate manner, AB-MIML locates potentially affected areas and provides better interpretability.

## 2. Related works

In this section, we first review relevant studies on machine learning-based diagnosis for periodontal diseases. We then introduce related research and applications of multi-instance multi-label learning and abductive learning paradigms.

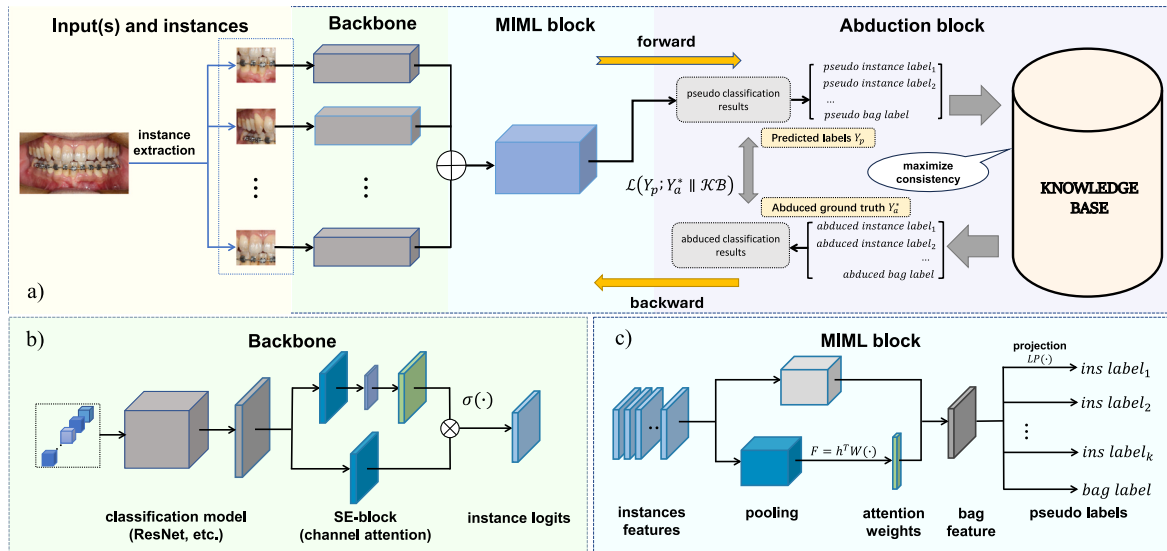


Fig. 2. The detailed illustration of our proposed Abductive Multi-Instance Multi-Label Learning (AB-MIML) framework for periodontal disease classification. Fig. 2 (b) and Fig. 2 (c) shows the details of the Backbone and MIML block respectively.

## 2.1. AI-assisted diagnosis for periodontal disease

With the popularity of artificial intelligence, AI-assisted diagnosis of periodontal diseases has received increasing attention from researchers. Early studies mainly center around exploring the probability of applying machine learning based approaches to the diagnosis process of periodontal diseases (Özden et al., 2015; Chen et al., 2018; Feres et al., 2018; Hung et al., 2019; Na et al., 2020; Farhadian et al., 2020). The attention of these studies primarily lies in the application of traditional machine learning technologies (Özden et al., 2015), which are simple yet easy to implement. While these methods have shown great promise and applicability in early AI-assisted periodontitis diagnosis, their effectiveness seems to be limited when handling high-resolution images and real-world medical scenarios.

With the development of deep learning, recent studies mainly utilize the advantages of deep models in image tasks to construct auxiliary diagnosis systems. These studies can be broadly categorized into two groups: detection-based and others. The latter methods primarily utilize the capabilities of established CNN models in image classification to directly create an end-to-end classification task for periodontal images (Imangaliyev et al., 2016; Johari et al., 2017; Joo et al., 2019; Mei et al., 2023). However, since periodontal disease is primarily reflected in the subtle details of textures, the accuracy of these methods is often not convincing. Therefore, the detection-based approach is commonly used, which is based on a two-stage paradigm: first, possible diseased patches are detected and classified from the images, and then further classification is performed based on the patch-level results to determine the labels of the whole images. The study in Chang et al. (2020) used RCNN to classify oral X-rays of a group of periodontitis patients to differentiate five different types of periodontal diseases. Another study in Li et al. (2021) used RCNN to classify the degree of gingivitis lesions. Compared to RCNN, YOLO-like models feature more complex structures and deliver superior results. One study in Ding et al. (2021) used YOLO-like models to detect and classify periodontal disease in panoramic images. After training the model, the results show that it demonstrates better classification accuracy than RCNN. Another study in ÖZİÇ et al. (2023) used YOLO to detect and classify bone loss areas in panoramic radiographs, and the results showed that YOLO could accurately identify and localizing these regions. Overall, AI-assisted methods have made significant advancements and demonstrated their value in diagnosing periodontal diseases. However, they still face certain limitations when it comes to effectively harnessing prior expert knowledge and applying it in real-world situations.

## 2.2. Multi-instance multi-label learning

In traditional supervised learning settings, an object is represented by only one instance and associated with a single label. This learning paradigm is widely applied in various fields, including resource exploration and traffic modeling (Zhang et al., 2020; Fouzi et al., 2021; Harrou et al., 2022). However, in real-world scenarios, an object is usually decomposed into multiple slices or instances (Wei et al., 2021), and associated with multiple class labels (Ye et al., 2016). Multi-instance multi-label learning (MIML) is a paradigm proposed for modeling such complicated correspondences between objects and their labels (Zhou and Zhang, 2006). As stated in Zhou et al. (2012), the multi-learning frameworks result from the ambiguities in representing complicated real-world objects. In recent years, the MIML paradigm has gained tremendous attention, and many learning approaches for MIML have been developed. Following its initial formalization, a generative model for MIML was proposed by Yang et al. (2009). Other feasible methods for MIML, including maximum margin approaches (Zhang and Zhou, 2008), neural networks (Zhang and Wang, 2009), SVM (Nguyen, 2010) and nearest neighbor methods (Zhang, 2010), were then proposed based on machine learning techniques. There were also some studies trying to extend the MIML paradigm to novel settings, including relation extraction (Surdeanu et al., 2012), positive unlabeled (PU) learning (Yang et al., 2013), novel class discovery (Zhu et al., 2017) and large-scaled application (Huang et al., 2019). The machine learning based MIML approaches was applied to various scenarios, like biodiversity assessment (Briggs et al., 2016) and medical imaging (Mercan et al., 2017). In addition, deep learning methods have also been applied to MIML (Feng and Zhou, 2017). Several popular deep learning methods for MIML have been successfully applied across various fields, including image classification and video capturing (Song et al., 2018; Ji and Wang, 2021). The MIML framework has emerged as an innovative approach to addressing complex real-world challenges, encompassing both input and output dimensions. It has been proved effective in a wide range of contexts and has been successfully integrated into numerous intelligent applications.

## 2.3. Abductive learning

Since the rise of artificial intelligence, symbolism and connectionism have been two major paradigms in the field, which have developed independently of each other for years. However, cognitive studies (Solso

- Examples for domain knowledge of bag-instance correspondence interpretation:

$$\neg \text{swelling}(X) \wedge \neg \text{recession}(X) \leftarrow \text{is\_bag}(X) \wedge \text{is\_healthy}(X)$$

$$\neg \text{swelling}(Y) \wedge \neg \text{recession}(Y) \leftarrow \text{instance\_of}(X, Y) \wedge \text{is\_healthy}(X)$$

- Examples for domain knowledge of professional diagnosis of periodontal disease:

$$\neg \text{recession}(Z) \leftarrow \text{center}(Y, Z) \wedge \neg \text{recession}(Y)$$

$$\text{recession}(Y) \leftarrow \text{center}(Y, Z) \wedge \text{center}(Y, Z') \wedge \text{recession}(Z) \wedge \text{recession}(Z')$$

Fig. 3. Sample formulas in the knowledge base.

et al., 2005) have shown that human cognitive processes often involve a combination of both. This discovery inspires us that the next generation of artificial intelligence calls for the integration of data-driven machine learning and knowledge-driven logic reasoning. ABductive Learning (ABL) (Zhou, 2019; Zhou and Huang, 2022) is a novel framework which leverages machine learning with logical reasoning in a *balanced-loop* enabling them to work together in a mutually beneficial way. Specifically, the machine learning classifier learns to make predictions on unseen data, which are then converted into pseudo-grounded facts; the abductive reasoning process (Magnani, 2009) is then applied to maximize the consistency between pseudo-grounded facts and prior knowledge. In recent years, multiple researches have worked on improving this novel paradigm. Some successful trials include handwriting recognition (Dai and Zhou, 2017), stroke evaluation (Wang et al., 2021) and theft judicial sentencing prediction (Huang et al., 2020). Meanwhile, other studies have combined the paradigm with new settings in machine learning, such as new concepts discovery (Huang et al., 2023a), inference acceleration (Huang et al., 2021) and knowledge graphs (Huang et al., 2023b). Overall, the abductive learning paradigm has made feasible explorations in the combination of machine learning with logical reasoning, and has been proved practical in several applications.

### 3. Methodology

#### 3.1. Problem formulation

Before diving into the details of our proposed method, we first provide a formalization of the problem. To classify and diagnose common periodontal diseases, we collect a dataset in the following form:  $D = \{(X_1, Y_1), \dots, (X_m, Y_m)\}$ .  $X_i \subseteq \mathcal{X}$  corresponds to the intraoral images of patients, and  $Y_i \subseteq \mathcal{Y} \subseteq [C] = \{1, \dots, C\}$  are the associated labels. In periodontal disease detection, we have  $C = 4$  for the four classes, namely, periodontal health (PH), gingivitis (GI), mild to moderate periodontitis (MMP), and severe periodontitis (SP). The goal is to learn a mapping  $f : \mathcal{X} \rightarrow \mathcal{Y}$  from  $D$ , which is able to make accurate predictions for a new intraoral image sampled from the same distribution of  $D$ .

#### 3.2. Motivation and overall architecture

Unlike the recognition of natural images, the diagnosis of periodontal diseases depends heavily on the identification of specific intraoral regions in  $X_i$ , and the diagnostic results of them collectively form the criteria for diagnosing gingivitis or periodontitis, as well as their severity. Furthermore, the relationships among disease categories should not be regarded as completely parallel. Rather, they possess a hierarchical order. For example, if we treat MMP, GI and PH as completely independent, then misclassifying MMP as GI will be considered completely erroneous, resulting in a loss equivalent to misclassifying it as PH. However, in reality, when the model misclassifies MMP as GI, it merely fails to recognize the recession patches, while successfully identifying the swelling patches. This situation is fundamentally different from

the error of classifying it as PH. Thus, treating disease categories as parallel not only undermines the independence of disease labels but also fails to align with the actual diagnostic process. Apparently, the standard mapping function  $f$ , constructing a “one-to-one” mapping from the input images to the labels, fail to capture the complicated correspondence between the regions and their labels.

To tackle this problem, we propose to transform the diagnosis task into a multi-instance multi-label setting. In the concerned problem, each intraoral image  $X_i$  is treated as a “bag”, which contains multiple local patches  $\{x_{i1}, x_{i2}, \dots, x_{i, n_i}\}$  as “instances”. Based on the expert knowledge, the four classes can be described by several aspects, i.e., gingival swelling or not, recession or not, and severe recession or not. Then the labels  $y_i$  can be transformed into sets of sub-labels  $\{y_{i1}, y_{i2}, y_{i3}\}$ , where the four classes have labels  $\{0, 0, 0\}$ ,  $\{1, 0, 0\}$ ,  $\{1, 1, 0\}$ , and  $\{1, 0, 1\}$ , respectively. The final label of an image can be obtained as follows: First, instance-level representations are aggregated to generate the bag-level predictions for the sub-labels; then the sub-labels are translated into the corresponding disease labels.

Although MIML methods can enhance modeling accuracy, they also introduce additional challenges for training the classification model. In fact, we have observed that periodontal images obtained from medical examinations typically possess a standardized structure, and doctors often have prior knowledge of common disease locations and adjacent information when making diagnostic assessments. Hence, to further take advantage of expert knowledge in the training process, we incorporate abductive learning paradigm to perform abductive reasoning on the predicted pseudo-labels with the knowledge base  $\mathcal{KB}$ , which contains prior domain knowledge expressed in first-order logic. The assistance of  $\mathcal{KB}$  helps to rectify the predictions for sub-labels, thereby improving the predictive capability of  $f$ . We propose the ABductive Multi-Instance Multi-Label learning (AB-MIML) method as follows:

- **AB-MIML:** Given a knowledge base  $\mathcal{KB}$  and a dataset  $\{(X_1, Y_1), (X_2, Y_2), \dots, (X_m, Y_m)\}$ , where  $X_i \subseteq \mathcal{X}$  is a set of instances  $\{x_{i1}, x_{i2}, \dots, x_{i, n_i}\}$  with  $x_{ij} \in \mathcal{X}$  ( $j = 1, 2, \dots, n_i$ ), and  $Y_i \subseteq \mathcal{Y}$  are the set of labels  $\{y_{i1}, y_{i2}, \dots, y_{i, l}\}$  with  $y_{ik} \in \mathcal{Y}$  ( $k = 1, 2, \dots, l$ ). Here  $n_i$  denotes the number of instances in  $X_i$ , and  $l$  the number of decomposed labels. AB-MIML aims to learn a function  $f : \mathcal{X} \rightarrow \mathcal{Y}$ , which satisfies the constraint  $\mathcal{KB} \models \mathcal{O}(f)$ , where  $\mathcal{O}(f)$  represents the facts grounded by  $f$ , and  $\models$  stands for logical entailment, meaning that  $\mathcal{O}(f)$  is consistent with  $\mathcal{KB}$ .

The overall architecture is shown in Fig. 2. The original image is divided into small patches after standard pre-processing, and sent to the backbone for automated feature extraction. Then multiple feature vectors of original instances are forwarded to the MIML block, where they are aggregated and mapped to the predicted labels. To effectively utilize professional knowledge and achieve high accuracy under coarse-grained annotations, we generate corresponding bag-level and instance-level pseudo-labels at this stage. These pseudo-labels are then sent to the knowledge base for abductive reasoning and correction. After the reasoning step, the labels align more closely with ground facts. We treat the abducted labels as ground truth, calculate the loss, and update the model using gradient-based methods.

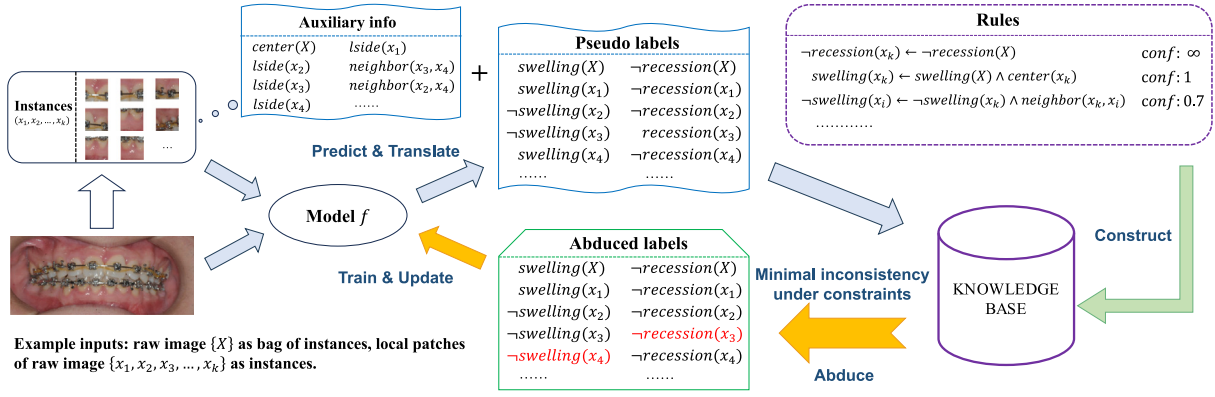


Fig. 4. The detailed illustration of the abduction process. The pseudo labels are first translated into logical variables and facts. Then we try to maximize the consistency between pseudo labels and the knowledge base by modifying some of the labels under cost constraints. For example, according to the logical rules, the bag-level annotation “no recession” means that all instances shall not show signs of recession, so the “recession” label of  $x_3$  is modified. Also, according to the rules, since the instances around the center of the image are more likely to show signs of affection, as well as all neighbors show no signs of swelling, the label “swelling” of  $x_4$  is modified, which is consistent with the ground truth.

### 3.3. Backbone

The backbone of our proposed method mainly consists of a classification model and a channel attention block. Since the diagnosis of periodontal diseases is essentially an image classification problem, almost all image classification models can be effectively applied in our approach, such as ResNet (He et al., 2016), ResNeXt (Xie et al., 2017) and ViT (Dosovitskiy et al., 2021). We will demonstrate later in experiments that our framework shows effective improvements under these backbones. Since the input is a three-channel color image, we add a channel attention module to help locate valuable channel information. The channel attention block is inspired by Hu et al. (2018), which can be described as follows:

$$g_c = \tanh(f_c * W_S) * W_E$$

$$h_c = f_c \cdot g_c$$
(1)

The symbol  $*$  denotes the matrix product, while  $\cdot$  represents the element-wise multiplication between two tensors. The  $f_c$  and  $h_c$  represent the input and output of channel attention block, in the shape of  $B \times 1 \times N$ , where  $B$  represents the batch size, and  $N$  represents the length of feature vector, or the number of output channels. The input feature is firstly squeezed into a low-dimension feature space and activated by  $\tanh$  function. Then the squeezed feature is restored to the same dimension as the input and used as the weight vector for the original input channels. Since the dimension of the intermediate feature space is only 1/16 to 1/32 of the original space, the squeezing operation can reduce the number of parameters used for weight learning. Such architecture can be viewed as a compression and reconstruction process of the input, enhanced by non-linear components through activation functions, hence better reflecting the importance of each dimension.  $W_S$  and  $W_E$  represent the parameter matrix of the squeezing and reconstructing process, with dimension  $N \times M$  and  $M \times N$ , where  $M$  represents the dimension of the intermediate feature space. In the end, the weight vector is multiplied with the original input vector to obtain the feature vector after channel reweighting.

### 3.4. MIML block

The MIML block is proposed to model the complex correspondence between input bags-of-instances and output labels. The backbone embeds the patches of each input image into a unified feature space while retaining the semantic connections among them. Therefore, the MIML module is designed to learn a function that maps the input feature vectors to the disease labels we need. In order to make our method more flexible, we use a learnable neural network to model the

correspondence between inputs and outputs. When generating multiple labels, the classifiers share the same embeddings of inputs. We leverage the attention-based multi-instance learning block (Ilse et al., 2018) to learn the aggregation process:

$$g_m^T = \text{Softmax}(\sigma(f_m * W_{att}))$$

$$h_m = g_m * f_m$$
(2)

Here,  $f_m$  and  $h_m$  represent the input and output of multi-instance block, in the shape of  $B \times K \times N$  and  $B \times 1 \times N$ , where  $K$  represents the number of instances.  $g_m$  represents the attention weight vector in the shape of  $1 \times K$ .  $\sigma$  represents the  $\tanh$  activation function. The input feature vectors are projected into a one-dimension constant vector, and then transferred into a weight vector summing up to one. The reweighted feature  $h_m$  is sent to multiple linear layers to generate pseudo-labels, which are then sent to the abduction block.

### 3.5. Abduction block

In abductive learning paradigm, the machine learning model learns to recognize primitive logic facts from raw data, while logical abduction utilizes symbolic domain knowledge to correct the misperceived facts. This abductive reasoning process enhances the performance of machine learning models. It can be formalized as follows (Zhou, 2019):

$$\{x_1, \dots, x_m\}, f \triangleright \mathcal{O}$$

$$\text{s.t.} \quad \mathcal{KB} \models \mathcal{O}, \text{ or}$$

$$\mathcal{KB} \models \Delta(\mathcal{O}), \quad f \leftarrow \psi(f, \Delta(\mathcal{O}))$$
(3)

This formalization is described in the form of logical reasoning, and we can translate like this: given dataset  $\{x_1, \dots, x_m\}$  and knowledge base  $\mathcal{KB}$ , the aim of it is to train a function  $f$  that makes predictions on unseen data, where the facts grounded by  $f$  are consistent with the knowledge base, or can get better consistency with knowledge base after adjustment  $\Delta$ . In the abduction block, the output pseudo-labels of the machine learning part are converted into logical terms, which are used for abductive reasoning of the knowledge base. This process mainly involves two key components:

**Knowledge base.** For the knowledge base, we transform the structural information of intraoral images and the professional knowledge of doctors into logical formulas. Some examples are shown in Fig. 3. The first two samples show the rules of bag-instance correspondence. For instance, if the bag-level annotation indicates that the patient is healthy, then none of the instance-level labels should indicate any affected factors. The last two samples shows the rules of the domain knowledge of professional diagnosis, indicating the potential affected

areas with high probabilities. Expert knowledge can identify regions where diseases are likely to occur and the relationships between different diseases, allowing for label corrections within the neighborhood of instances. Our knowledge base includes approximately 17 logical formulas that encompass the primary structural information and criteria for diagnosing periodontal disease. We consider these logical formulas as ground-truth prior knowledge in our abduction process.

**Abduction process.** For abductive reasoning, we first align the pseudo-labels with the logical variables in the knowledge base. Then we verify the consistency between input logical variables and the knowledge base. The whole abduction process is illustrated in Fig. 4. Briefly, once the truth values of logical variables are assigned, the values of logical formulas are determined. If the value of a logical formula is true, it is viewed as **satisfied** under given inputs, otherwise it is **unsatisfied**. After the variable values are given, the number of satisfied formulas in the knowledge base is defined as the consistency between the input and the knowledge base. If the input variables are not completely consistent with the knowledge base, we attempt to modify the value of inputs according to the ground-truth knowledge base within the limit of modification cost to maximize consistency. Given knowledge base  $\mathcal{KB}$  and function  $f$ , the optimization objective is shown as:

$$\begin{aligned} \max_{\Delta(\mathcal{O})} \quad & \text{Con}[\psi(f, \Delta(\mathcal{O}))] - \text{Con}[f] \\ \text{s.t.} \quad & \text{Cost}[\Delta(\mathcal{O})] \leq T \end{aligned} \quad (4)$$

Here,  $f$  represents the function that maps raw inputs to the predicted pseudo-labels. We want to maximize the difference between the consistency of abduced  $f$  and the original one with the help of the knowledge base. The operation  $\text{Con}(\cdot)$  measures the consistency of the function with the given knowledge base  $\mathcal{KB}$ . The operation  $\text{Cost}(\cdot)$  measures the cost of adjusting the facts (e.g.,  $\mathcal{O}$ ) grounded by  $f$ . It is important to note that in the process of abductive reasoning, each step of modifying logical variables incurs a certain cost. This cost is associated with the number of label changes or the increase in inconsistency before and after these changes. Since for any set of logical formulas, there exists a specific combination of truth values for the logical variables that maximizes the satisfiability of the formulas. However, such a global optimal combination is meaningless for most practical samples. So the reasoning process needs to be done under cost constraints so as to avoid converging to the global optimal. Considering the limitations imposed by cost in the reasoning process, not all rules can be satisfied. Some rules derive from the definition of the problem, and violation of these rules may lead to serious consequences. Other rules derive from empirical induction, and their violation may not have a significant impact. Therefore, we can divide the rules in the knowledge base into soft and hard parts. The hard rules have a significant impact on the problem's solution and must be satisfied. The remaining rules can be selectively satisfied based on the optimization algorithm. Given knowledge base  $\mathcal{KB}$  and function  $f$ , the problem with the introduction of soft-and-hard rules can be formalized as:

$$\begin{aligned} \max_{\Delta(\mathcal{O})} \quad & \text{Con}[\psi(f, \Delta(\mathcal{O}))] - \text{Con}[f] \\ \text{s.t.} \quad & \mathcal{KB} \models \mathcal{O}^+, \\ & \text{Cost}[\Delta(\mathcal{O})] \leq T \end{aligned} \quad (5)$$

The choice of optimization objectives relies on the problem settings. In practice, the bag-level labels annotated by doctors can be viewed as ground-truth, so the associated rules should be completely satisfied. Then we can apply the soft-and-hard paradigm to ensure the bag-level labels are not modified. However, there are also certain circumstances in which the given annotated images are noisy or partially mislabeled. In this case, the original or the weighted form of optimization objectives are more suitable. By solving the above optimization problem, we obtain abduced labels that are *more consistent with prior knowledge and ground-truth*, which are then returned to the perception modules to calculate the training loss.

### 3.6. Overall optimization objective

According to the aforementioned details of our proposed method, the overall optimization objective can be described as follows:

$$\begin{aligned} \min_f \quad & - \sum_{n,c} \log P(Y_{n,c} | X_n) - \alpha \sum_{n,c,i} \log P(y_{n,c}^i | x_n^i) \\ & - \beta \sum_i \text{Con}(\Delta(Y_n), \mathcal{KB}_\theta) \end{aligned} \quad (6)$$

This optimization objective consists of two parts. The first part is the cross entropy loss measuring the quality of classification results in perception, and the second part is the inconsistency between pseudo-labels and knowledge base in reasoning. To achieve good classification results, we calculate the loss at both bag-level and instance-level, in order to make predictions on the whole images and local patches correctly at the same time. The letter  $i$  in (6) represents the number of instances contained in a bag. The ground-truth labels for bag-level annotation are gained from the doctors, and the ground-truth for instance-level classification is obtained by abduction. Moreover, we add a balancing coefficient  $\alpha = \alpha(r)$  to balance the bag-level and instance-level objectives. For the latter one, we simply try to minimize the inconsistency of each predicted sample with the knowledge base  $\mathcal{KB}_\theta$ , where  $\theta$  represents the variables of  $\mathcal{KB}$ . During training, the framework needs to optimize two objectives simultaneously. On the model side, we need to minimize the value of the loss function by gradient-based methods. On the logic side, we seek to minimize the inconsistency between the labels and the knowledge base while adhering to specific cost constraints. The abduction process can be viewed as a variant of the MAX-SAT problem, which is recognized as NP-hard. When the problem size is small, the abduction process can be solved using traversal methods. However, as the problem size increases, optimization can instead be approached using heuristic methods, depending on the specific problem size and the strategies employed by the solvers (Li and Manyà, 2021).

## 4. Experiments

In this section, we will first introduce the dataset and experimental setups. Then we will validate our approach by comparing it with non-abduction classification methods, as well as detection-based methods. Then we will discuss the performance of our proposed approach under various settings to demonstrate its robustness. Finally, we will illustrate its effectiveness in locating possible diseased areas, which aligns closely with determinations made by human doctors.

### 4.1. Introduction to datasets

The datasets used in our study were collected and obtained from Nanjing Stomatological Hospital. Specifically, we gathered original images of all patients treated in the orthodontic department from 2017 to 2020. We extracted a selection of images from the original sequences and invited experts from the periodontal department to annotate them into four classes: periodontal health (PH), gingivitis (GI), mild to moderate periodontitis (MMP), and severe periodontitis (SP). Detailed information about the dataset is presented in Table 1. The statistics in the table are compiled at the image level. For example, for the PH class, we have a training set consisting of 1,531 images. We divide the raw dataset into training and testing parts by a random split. The test sets are organized by their corresponding year in order to prove that our method is robust during the time span of our research. All the raw data is labeled at the image level, and does not include any annotations for localizing affected patches. In other words, during the training process, we only have access to bag-level labels, while instance-level labels are generated through abductive reasoning based on their pseudo-labels and the corresponding prior knowledge.

All raw data were taken by digital cameras, but due to the long span of collection, there may be differences in shooting equipment and raw

**Table 1**

Detailed information of datasets. The training set is collected from the whole time span, while the test sets are organized by year.

| Mission & Year | Train | Test |      |      |      |
|----------------|-------|------|------|------|------|
|                |       | 2017 | 2018 | 2019 | 2020 |
| PH             | 1531  | 122  | 167  | 159  | 160  |
| GI             | 1573  | 202  | 176  | 215  | 227  |
| MMP            | 1575  | 189  | 183  | 189  | 183  |
| SP             | 295   | 28   | 30   | 27   | 23   |

data size. We employ several traditional image processing algorithms to identify the contours of the gums in each image and crop the images accordingly, ensuring that all input images focus on the patient's periodontal areas. The resolution of the cropped images ranges from  $3 \times 800 \times 1200$  (in the form of  $C \times H \times W$ ) to  $3 \times 4000 \times 6000$  pixels. For ease of processing, the original images in the non-MIML settings are uniformly resized to  $3 \times 384 \times 384$  pixels. In the MIML settings, we decompose each image into 10 patches based on the contours obtained during preprocessing, with each patch resized to  $3 \times 128 \times 128$  pixels. Some multi-instance examples are shown in Fig. 5. This approach is adopted to balance any potential differences arising from variations in image clarity between the two groups of methods.

#### 4.2. Experiments with diverse backbones

As noted in the related works section, there are primarily two types of existing classification methods for periodontal diseases based on deep learning models: One is detection-based, while the other is not. Since our proposed method does not involve region detection techniques, this section will first present comprehensive experiments comparing our methods with existing disease classification approaches. Our experiments involve several state-of-the-art (SOTA) and commonly used image classification models for periodontal diseases, including VGG (Alotaibi et al., 2022), ResNet (Moran et al., 2020), ResNeXt (Fariza et al., 2024) and ViT (Dujic et al., 2023) to demonstrate that our proposed method can be applied to most existing models and enhance their performance in periodontal image classification tasks. During each training epoch, we reserve approximately one-tenth of the training dataset to validate the performance of the model. If the model's performance does not show significant improvement after more than five epochs, we will cease the training process. The detailed experimental results are presented in Table 2. In the table, the final results are organized based on the backbone of the algorithm and the year of the test set. The experiments in Table 2 are all conducted under the same pretrained setting. The **MIML** mark means that the experiment is conducted under the MIML setting, and the **ABMIML** mark means that the experiment is assisted by our proposed approach. In one set of experiments in Table 2, we examine the backbone with and without the assistance of our proposed methods. For each group of experiments, we indicate the best results with an \*, while the global optimal results are highlighted in bold. Based on the experimental results, we can safely draw the following conclusions:

**Our proposed framework achieves the best performance beyond baseline models in almost all metrics.** To demonstrate the superiority of our framework, we employ diverse performance measures, such as accuracy (ACC), macro-precision (mPRE), macro-recall/sensitivity (mSEN), macro-specificity (mSPE), macro-F1-score (mF1), and macro-AUC (mAUC). By vertically comparing experiments within each group, it becomes evident that our AB-MIML approach demonstrates significant advantages in addressing periodontal disease classification problems compared to the direct classification of original images. By comparing the base groups to the \*-MIML groups, we can safely conclude that our modeling approach for this issue is more effective, revealing the intricate relationship between input images and their corresponding labels. Moreover, through the application of abductive reasoning, each

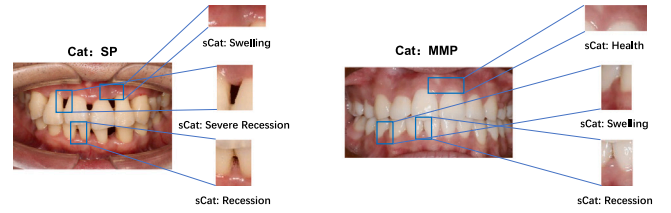


Fig. 5. Examples of multi-instance categories of periodontal diseases. Cat. is short for category, and sCat. stands for decomposed sub-labels.

backbone model has achieved performance measures surpassing the baselines, which is depicted by comparing the \*-MIML and \*-ABMIML groups. Such vertical comparisons can also serve as ablative experiments, demonstrating the effectiveness of our AB-MIML framework in both problem modeling and prior knowledge leveraging.

**Our proposed method achieves strong stability across different periods of time.** Due to the long time span of the periodontal images selected in this study, we organize the test set by year to verify the robustness of our proposed method. We test both the baseline method and our approach on the test sets from different years. We take the experimental group with ViT as the backbone as an example, and Fig. 6 shows the fluctuations in accuracy of different models on the test sets from 2017 to 2020. The final experimental results indicate that our method exhibits minimal fluctuation in performance across the various test sets, with overall accuracy ranging from 84% to 85%, which is approximately 5% higher than the baseline method. These results suggest that our method achieves state-of-the-art accuracy across datasets from different years, demonstrating its robustness over time.

#### 4.3. Experiments with detection models

In this section, to validate the effectiveness of our proposed approach, we design an experiment to compare our method with SOTA periodontal disease diagnosis approaches based on detection models. Our investigation reveals that current research on auxiliary diagnosis of periodontal diseases primarily relies on two types of object detection models: region-based models, such as RCNN and Faster-RCNN (Alalharith et al., 2020; Kong et al., 2023; Chen et al., 2023), and YOLO models, which can directly output bounding boxes and classification labels from images (Muresan et al., 2020; Farook et al., 2023; Cholan et al., 2023). A review of these studies can be found in Section 2.1. Since most of these researches have not made its code and datasets publicly available, to ensure a fair comparison, we will evaluate our method against the performance of these backbone models,

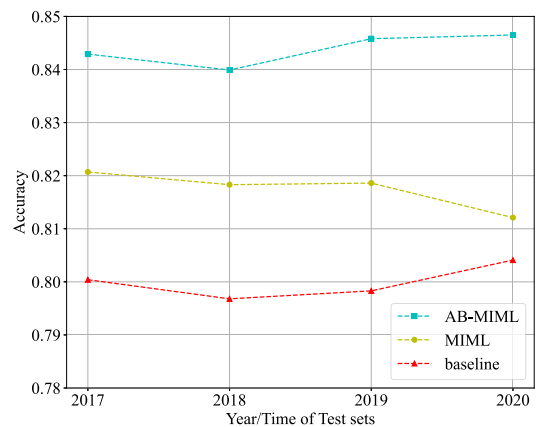


Fig. 6. An illustration of average performance of different methods across test data from 2017 to 2020.

**Table 2**

Experimental results of different methods and models evaluated on several main metrics from 2017 to 2020. We assume that the labels of the original images are ground truth. The results are organized by groups of backbones and times.

| Methods & Models | Metrics & Results |         |         |         |         |         |         |         |         |         |         |         |
|------------------|-------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
|                  | 2017              |         |         |         |         |         | 2018    |         |         |         |         |         |
|                  | ACC               | mPRE    | mSEN    | mSPE    | mF1     | mAUC    | ACC     | mPRE    | mSEN    | mSPE    | mF1     | mF1     |
| VGG16            | 0.6987            | 0.7552  | 0.7297  | 0.8903  | 0.7374  | 0.8570  | 0.7104  | 0.7834  | 0.7428  | 0.8946  | 0.7584  | 0.8747  |
| VGG16-MIML       | 0.7246            | 0.7753  | 0.7592  | 0.9000  | 0.7625  | 0.8824  | 0.7320  | 0.8026  | 0.7601  | 0.9026  | 0.7759  | 0.8625  |
| VGG16-ABMIML     | 0.7320*           | 0.7805* | 0.7797* | 0.9026* | 0.7763* | 0.8874* | 0.7446* | 0.8118* | 0.7697* | 0.9071* | 0.7859* | 0.8920* |
| ResNet18         | 0.7782            | 0.8160  | 0.8110  | 0.9188  | 0.8116  | 0.9284  | 0.7770  | 0.8354  | 0.7961  | 0.9188  | 0.8109  | 0.9259  |
| ResNet18-MIML    | 0.7911            | 0.8272  | 0.8254  | 0.9239  | 0.8233  | 0.9407  | 0.7896  | 0.8451  | 0.8126  | 0.9234  | 0.8247  | 0.9396  |
| ResNet18-ABMIML  | 0.8133*           | 0.8426* | 0.8449* | 0.9325* | 0.8406* | 0.9446* | 0.8094* | 0.8595* | 0.8349* | 0.9306* | 0.8437* | 0.9462* |
| ResNet50         | 0.7930            | 0.8282  | 0.8189  | 0.9243  | 0.8209  | 0.9422  | 0.7968  | 0.8515  | 0.8252  | 0.9260  | 0.8341  | 0.9451  |
| ResNet50-MIML    | 0.8152            | 0.8449  | 0.8460  | 0.9329  | 0.8425  | 0.9477  | 0.8129  | 0.8605  | 0.8380  | 0.9319  | 0.8463  | 0.9462  |
| ResNet50-ABMIML  | 0.8318*           | 0.8577* | 0.8573* | 0.9384* | 0.8558* | 0.9510* | 0.8291* | 0.8713* | 0.8506* | 0.9378* | 0.8587* | 0.9524* |
| ResNeXt          | 0.7985            | 0.8328  | 0.8243  | 0.9264  | 0.8255  | 0.9407  | 0.7932  | 0.8462  | 0.8222  | 0.9248  | 0.8310  | 0.9410  |
| ResNeXt-MIML     | 0.8189            | 0.8495  | 0.8380  | 0.9331  | 0.8418  | 0.9522  | 0.8201  | 0.8678  | 0.8439  | 0.9345  | 0.8521  | 0.9498  |
| ResNeXt-ABMIML   | 0.8336*           | 0.8599* | 0.8714* | 0.9389* | 0.8642* | 0.9584* | 0.8381* | 0.8776* | 0.8575* | 0.9410* | 0.8655* | 0.9610* |
| ViT              | 0.8004            | 0.8349  | 0.8452  | 0.9270  | 0.8375  | 0.9436  | 0.7968  | 0.8481  | 0.8249  | 0.9261  | 0.8337  | 0.9447  |
| ViT-MIML         | 0.8207            | 0.8501  | 0.8607  | 0.9344  | 0.8534  | 0.9541  | 0.8183  | 0.8618  | 0.8559  | 0.9340  | 0.8572  | 0.9510  |
| ViT-ABMIML       | 0.8429*           | 0.8741* | 0.8777* | 0.9421* | 0.8748* | 0.9657* | 0.8399* | 0.8789* | 0.8728* | 0.9417* | 0.8740* | 0.9625* |

| Methods & Models | Metrics & Results |         |         |         |         |         |         |         |         |         |         |         |
|------------------|-------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
|                  | 2019              |         |         |         |         |         | 2020    |         |         |         |         |         |
|                  | ACC               | mPRE    | mSEN    | mSPE    | mF1     | mAUC    | ACC     | mPRE    | mSEN    | mSPE    | mF1     | mAUC    |
| VGG16            | 0.7360            | 0.7883  | 0.7752  | 0.9028  | 0.7793  | 0.8886  | 0.7239  | 0.7765  | 0.7443  | 0.8973  | 0.7570  | 0.8870  |
| VGG16-MIML       | 0.7411            | 0.7925  | 0.7786  | 0.9046  | 0.7833  | 0.8961  | 0.7391  | 0.7880  | 0.7557  | 0.9030  | 0.7685  | 0.8954  |
| VGG16-ABMIML     | 0.7530*           | 0.8009* | 0.7872* | 0.9089* | 0.7923* | 0.9014* | 0.7542* | 0.8007* | 0.7768* | 0.9086* | 0.7863* | 0.9012* |
| ResNet18         | 0.7712            | 0.8153  | 0.8031  | 0.9158  | 0.8066  | 0.9356  | 0.7690  | 0.8122  | 0.7890  | 0.9142  | 0.7979  | 0.9342  |
| ResNet18-MIML    | 0.7949            | 0.8338  | 0.8210  | 0.9244  | 0.8252  | 0.9462  | 0.7909  | 0.8296  | 0.8041  | 0.9222  | 0.8146  | 0.9464  |
| ResNet18-ABMIML  | 0.8068*           | 0.8426* | 0.8458* | 0.9289* | 0.8426* | 0.9496* | 0.8044* | 0.8394* | 0.8249* | 0.9273* | 0.8305* | 0.9514* |
| ResNet50         | 0.7932            | 0.8317  | 0.8270  | 0.9247  | 0.8267  | 0.9401  | 0.7926  | 0.8312  | 0.8284  | 0.9233  | 0.8270  | 0.9412  |
| ResNet50-MIML    | 0.8102            | 0.8451  | 0.8573  | 0.9312  | 0.8462  | 0.9479  | 0.8094  | 0.8438  | 0.8389  | 0.9293  | 0.8398  | 0.9486  |
| ResNet50-ABMIML  | 0.8288*           | 0.8598* | 0.8701* | 0.9378* | 0.8625* | 0.9537* | 0.8280* | 0.8568* | 0.8547* | 0.9366* | 0.8538* | 0.9555* |
| ResNeXt          | 0.7966            | 0.8344  | 0.8293  | 0.9259  | 0.8294  | 0.9398  | 0.7943  | 0.8315  | 0.8316  | 0.9247  | 0.8275  | 0.9377  |
| ResNeXt-MIML     | 0.8186            | 0.8514  | 0.8483  | 0.9340  | 0.8470  | 0.9486  | 0.8044  | 0.8392  | 0.8382  | 0.9282  | 0.8354  | 0.9429  |
| ResNeXt-ABMIML   | 0.8356*           | 0.8651* | 0.8766* | 0.9401* | 0.8684* | 0.9597* | 0.8381* | 0.8639* | 0.8629* | 0.9407* | 0.8611* | 0.9588* |
| ViT              | 0.7983            | 0.8346  | 0.8406  | 0.9263  | 0.8352  | 0.9418  | 0.8061  | 0.8400  | 0.8409  | 0.9292  | 0.8362  | 0.9441  |
| ViT-MIML         | 0.8186            | 0.8510  | 0.8650  | 0.9340  | 0.8549  | 0.9477  | 0.8111  | 0.8439  | 0.8539  | 0.9311  | 0.8451  | 0.9462  |
| ViT-ABMIML       | 0.8458*           | 0.8727* | 0.8836* | 0.9436* | 0.8764* | 0.9640* | 0.8465* | 0.8707* | 0.8781* | 0.9438* | 0.8724* | 0.9690* |

including RCNN (Girshick et al., 2014), Faster-RCNN (Ren et al., 2015), YOLOv1 (Redmon et al., 2016) and YOLOv7 (Wang et al., 2023). For the mentioned models, we implement the well-established versions of them. Due to the substantial effort required from specialized doctors to annotate the target detection dataset, we are unable to complete the full annotation of the dataset mentioned in Section 4.2 within a short timeframe. So in this section, we prepare two datasets containing 1,200 and 2,000 finely-annotated images, respectively, which can be considered balanced in terms of categories. To ensure fairness, we re-evaluate all methods using these new datasets. During testing, we retain only those candidate boxes generated by the object detection model that have a confidence level greater than 0.5. The labels for the entire images are determined based on the types of candidate boxes present. For instance, if a candidate box indicates severe gum recession, the whole image will be classified as class SP. The results are illustrated in Table 3, and the global optimal results are highlighted by an asterisk. From the results, we can see that the AB-MIML method outperforms any object detection method in terms of accuracy for the periodontal disease diagnosis problem. To delve deeper into the reasons behind this, we provide the recall rates for each category. Notably, all models perform well in the SP class, which is probably due to the distinct differences in image representation compared to the other three classes. Moreover, as is shown in the table, it appears that the detection-based methods struggle to differentiate between the PH and GI classes. The results suggest that these models tend to misclassify GI as PH, indicating their inability to identify swelling areas in the images, which is probably due to the subtle texture differences between

swelling gums and healthy gums. Overall, while the detection-based methods perform well on some binary classification problems (e.g., PH vs. MMP and MMP vs. SP), they struggle to effectively differentiate between specific types of periodontal diseases in the four-class diagnosis problem at hand. In contrast, our proposed method outperforms the baseline detection-based methods across almost all metrics.

#### 4.4. Case study of robustness

To verify the robustness of our method in complex real-world scenarios, we conduct additional experiments to evaluate the performance of our proposed method. In this section, we will conduct experiments to validate the performance of our proposed method in both pretrained and non-pretrained scenarios, as well as in noisy environments. Additionally, due to the impact of class imbalance, we will provide an analysis of our method's performance on each individual category.

##### 4.4.1. The impact of pretraining

To validate that our proposed method performs well in both pretrained and non-pretrained scenarios, we conduct additional experiments in each context. In this section, the test sets are not organized by year; instead, they consist of a random selection of original images. For the SP class, the ratio is about 20%, and for other classes, the ratio is about 10%. The experimental results are presented in Table 4. In each group, we conduct experiments using both pretrained and non-pretrained models while keeping the backbone unchanged. The final results indicate that, with non-pretrained models, as the complexity of

**Table 3**

A brief exhibition of experimental results of our proposed method compared with the detection models. The backbone models in the experiment are all implemented using the well-established versions.

| Methods & Models | Metrics & Results |        |                   |                   |                    |                   |                  |        |                   |                   |                    |                   |
|------------------|-------------------|--------|-------------------|-------------------|--------------------|-------------------|------------------|--------|-------------------|-------------------|--------------------|-------------------|
|                  | SampleNum = 1200  |        |                   |                   |                    |                   | SampleNum = 2000 |        |                   |                   |                    |                   |
|                  | ACC               | mPRE   | SEN <sub>PH</sub> | SEN <sub>GI</sub> | SEN <sub>MMP</sub> | SEN <sub>SP</sub> | ACC              | mPRE   | SEN <sub>PH</sub> | SEN <sub>GI</sub> | SEN <sub>MMP</sub> | SEN <sub>SP</sub> |
| RCNN             | 0.7121            | 0.7418 | 0.9615            | 0.4151            | 0.6600             | 0.8372            | 0.7240           | 0.7510 | 0.9600            | 0.3846            | 0.7200             | 0.8750            |
| F-RCNN           | 0.6959            | 0.7220 | 0.9423            | 0.3725            | 0.6531             | 0.8333            | 0.7273           | 0.7510 | 0.9231            | 0.3962            | 0.7551             | 0.8636            |
| YOLOv1           | 0.7231            | 0.7463 | 0.8846            | 0.4118            | 0.7551             | 0.8605            | 0.7323           | 0.7568 | 0.9038            | 0.4340            | 0.7708             | 0.8444            |
| YOLOv7           | 0.7474            | 0.7730 | 0.9423            | 0.4400            | 0.7755             | 0.8372            | 0.7577           | 0.7797 | 0.8846            | 0.5000            | 0.7959             | 0.8605            |
| ResNet50-ABMIML  | 0.8041            | 0.8183 | 0.7692            | 0.7800            | 0.7959             | 0.8837            | 0.8196           | 0.8318 | 0.8077            | 0.8000            | 0.7959             | 0.8837            |
| ViT-ABMIML       | <b>0.8247*</b>    | 0.8339 | 0.8077            | 0.8000            | 0.7755             | 0.9302            | <b>0.8367*</b>   | 0.8445 | 0.8077            | 0.8400            | 0.7843             | 0.9302            |

**Table 4**

Experimental results of pretrained and non-pretrained models. To be brief, we select two representative backbones to demonstrate the effectiveness of our approach. The 'NaN' in the table indicates that the combination of methods can hardly converge to a feasible solution.

| Methods & Models | Metrics & Results |        |        |        |        |                |        |        |        |        |
|------------------|-------------------|--------|--------|--------|--------|----------------|--------|--------|--------|--------|
|                  | Pretrained        |        |        |        |        | Non-pretrained |        |        |        |        |
|                  | ACC               | mPRE   | mSEN   | mSPE   | mF1    | ACC            | mPRE   | mSEN   | mSPE   | mF1    |
| ResNet50         | 0.7906            | 0.8265 | 0.8194 | 0.9277 | 0.8214 | 0.7345         | 0.7554 | 0.7579 | 0.9024 | 0.7562 |
| ResNet50-MIML    | 0.8164            | 0.8389 | 0.8377 | 0.9288 | 0.8362 | NaN            | NaN    | NaN    | NaN    | NaN    |
| ResNet50-ABMIML  | 0.8308*           | 0.8584 | 0.8562 | 0.9342 | 0.8563 | 0.8178*        | 0.8372 | 0.8364 | 0.9162 | 0.8388 |
| ViT              | 0.8006            | 0.8368 | 0.8387 | 0.9273 | 0.8374 | 0.7534         | 0.7722 | 0.7753 | 0.9062 | 0.7748 |
| ViT-MIML         | 0.8165            | 0.8514 | 0.8615 | 0.9327 | 0.8560 | NaN            | NaN    | NaN    | NaN    | NaN    |
| ViT-ABMIML       | <b>0.8432*</b>    | 0.8683 | 0.8796 | 0.9415 | 0.8728 | <b>0.8207*</b> | 0.8427 | 0.8462 | 0.9356 | 0.8452 |

the training task increases (i.e., when the model's inputs shift from a single image to multiple related image patches and the correspondence between labels and inputs becomes more nuanced), there is a high likelihood of training failure for many commonly used models. These failed experiments are marked with NaN in the table, indicating that the model did not converge after 200 epochs, or that it only converged on one side (i.e., all inputs are assigned the same label). In such cases, abductive reasoning can leverage prior knowledge to help the model accurately identify the instances most likely to influence the final classification results, thereby facilitating convergence in an otherwise challenging training process. Additionally, with pretrained models, abductive reasoning can enhance model performance and support better decision-making. Therefore, from this perspective, integrating abductive reasoning into the classification process of periodontal diseases can be viewed as a form of model pretraining. Abductive learning provides additional information to the training process through prior expert knowledge, and constrains the model's gradient direction. As a result, abductive reasoning helps the model converge more quickly and improves the final classification performance.

#### 4.4.2. The impact of noisy datasets

To verify that our method still has good generality in complex real-world scenarios, we further explore the performance of our proposed approach under noisy settings. The results are presented in Table 5. In the experiment, we introduce different levels of noise to simulate potential errors in doctors' judgments that may occur in real situations. Such misclassifications in the labels can significantly impact the training process of the model, leading to the acquisition of erroneous knowledge. The noise rates  $\eta$  in our experiments are artificially manipulated. Specifically, we assume the original doctor annotation data to be error-free with a noise rate of 0. Subsequently, we randomly select samples from the training data of each class and shuffle them into other classes based on different noise rates. This shuffling process requires special consideration for the severe periodontitis category due to its significantly lower number of training samples compared to other categories. If samples from other categories are shuffled into the severe periodontitis category with equal probability, the noise rate in this category could exceed 50%, which makes it challenging for effective learning. To present the experimental results clearly, we conduct the experiments

using the ResNet-50 backbone model as a representative of the overall algorithmic performance. The experimental results demonstrate that irrespective of the noise rate, the overall trends remain consistent with those observed in a noise-free environment. Given the complexity of the MIML problem, the model struggles to converge without the instance-level constraints provided by the ABL method or proper initialization of model parameters using pre-trained models. However, under the same conditions, the ABL method significantly enhances the results. Notably, due to its incorporation of expert prior knowledge, the ABL method not only emphasizes the key components that contribute most to the final decision during training but also exhibits a degree of self-correction. As a result, the algorithm demonstrates considerable stability, even with increasing noise rates. As shown in the table, when the noise rate reaches 20%, the ABL algorithm experiences a performance loss of approximately 5%, while the non-ABL method suffers a performance loss exceeding 10%. This clearly highlights the advantageous performance of our AB-MIML framework in complex environments.

#### 4.4.3. Robustness across classes

As shown in Table 1, the dataset used in our experiment exhibits a notable class imbalance, with significantly fewer samples in the severe periodontitis category compared to the other categories. Although this data distribution aligns with the real-world incidence of disease severity, it also presents challenges for our training and experiments. These challenges are reflected in the insufficient number of training samples for the minor classes and the class imbalance in the final test set. Therefore, to validate the effectiveness of our proposed method, it is crucial to conduct a detailed analysis on both the baseline models and our proposed method across the four categories. To accomplish this, we design additional experiments using ResNet50 and ViT as the backbones to thoroughly analyze the performance of each method across different categories. The final experimental results are organized by class labels and illustrated in Table 6.

The experimental results reveal some counterintuitive findings. For instance, due to data scarcity, we expected the severe periodontitis category to be the hardest to train, resulting in lower accuracy and recall. However, in practice, classification results for severe periodontitis are consistently the best across all models. We believe this is because the image textures of the severe periodontitis category differ

**Table 5**

A brief overview of experimental results of different methods evaluated on several key metrics of test datasets. We assume that there is a certain ratio of noise in the bag-level annotation. The methods evaluated using a pretrained backbone are identified by the suffix “(P)”. The “NaN” in the table indicates that the model can hardly converge to a feasible solution under such settings.

| Methods & Models          | Metrics & Results |        |        |                |        |        |                |        |        |                |        |        |
|---------------------------|-------------------|--------|--------|----------------|--------|--------|----------------|--------|--------|----------------|--------|--------|
|                           | $\eta = 0\%$      |        |        | $\eta = 5\%$   |        |        | $\eta = 10\%$  |        |        | $\eta = 20\%$  |        |        |
|                           | ACC               | mSEN   | mF1    | ACC            | mSEN   | mF1    | ACC            | mSEN   | mF1    | ACC            | mSEN   | mF1    |
| <b>ResNet50</b>           | 0.7345            | 0.7579 | 0.7562 | 0.7300         | 0.7602 | 0.7696 | 0.7024         | 0.7464 | 0.7523 | 0.6605         | 0.7164 | 0.7153 |
| <b>ResNet50-MIML</b>      | NaN               | NaN    | NaN    | NaN            | NaN    | NaN    | NaN            | NaN    | NaN    | NaN            | NaN    | NaN    |
| <b>ResNet50(P)-MIML</b>   | 0.8164            | 0.8377 | 0.8362 | 0.8090         | 0.8254 | 0.8266 | 0.7588         | 0.7893 | 0.7830 | 0.7207         | 0.7536 | 0.7601 |
| <b>ResNet50-ABMIML</b>    | 0.8178            | 0.8364 | 0.8388 | 0.8120         | 0.8298 | 0.8314 | 0.7541         | 0.7842 | 0.7793 | 0.7157         | 0.7636 | 0.7614 |
| <b>ResNet50(P)-ABMIML</b> | <b>0.8308*</b>    | 0.8562 | 0.8563 | <b>0.8280*</b> | 0.8423 | 0.8507 | <b>0.7852*</b> | 0.8040 | 0.8121 | <b>0.7460*</b> | 0.7670 | 0.7614 |

**Table 6**

Experimental results for class-wise analysis. The models used in the experiment are all pretrained.

| Methods & Models       | Metrics & Results |                   |                   |                    |                   |                   |                   |                    |                   |  |
|------------------------|-------------------|-------------------|-------------------|--------------------|-------------------|-------------------|-------------------|--------------------|-------------------|--|
|                        | ACC               | PRE <sub>PH</sub> | PRE <sub>GI</sub> | PRE <sub>MMP</sub> | PRE <sub>SP</sub> | SEN <sub>PH</sub> | SEN <sub>GI</sub> | SEN <sub>MMP</sub> | SEN <sub>SP</sub> |  |
| <b>ResNet50</b>        | 0.7877            | 0.7290            | 0.7742            | 0.8559             | 0.8529            | 0.8721            | 0.6829            | 0.8112             | 0.8286            |  |
| <b>ResNet50-MIML</b>   | 0.8091            | 0.7480            | 0.7813            | 0.8819             | 0.9412            | 0.8676            | 0.7114            | 0.8394             | 0.9143            |  |
| <b>ResNet50-ABMIML</b> | 0.8344*           | 0.8000            | 0.7851            | 0.9043             | 0.9459            | 0.8767            | 0.7724            | 0.8353             | 1.0000            |  |
| <b>ViT</b>             | 0.7984            | 0.7364            | 0.7727            | 0.8697             | 0.9394            | 0.8676            | 0.6911            | 0.8313             | 0.8857            |  |
| <b>ViT-MIML</b>        | 0.8171            | 0.7610            | 0.7946            | 0.8782             | 0.9444            | 0.8721            | 0.7236            | 0.8394             | 0.9714            |  |
| <b>ViT-ABMIML</b>      | <b>0.8438*</b>    | 0.7935            | 0.8069            | 0.9106             | 1.0000            | 0.8950            | 0.7642            | 0.8594             | 0.9714            |  |

significantly from those of the other three categories. In images of severe periodontitis, the local patches frequently show significant root exposure, which is significantly different from the periodontal images in other categories (which can be seen from Fig. 1). Therefore, despite the limited amount of data, its final predictive performance is actually the highest among all categories. Furthermore, comparisons across each experimental group indicate that the AB-MIML method achieves significant improvements in accuracy and recall for nearly all categories, with the most substantial gains observed in the PH and SP categories. This may be attributed to our proposed method’s ability to impose greater loss on more dissimilar categories, thereby reducing the likelihood of erroneous classification results. Overall, the experiments demonstrate that our proposed method excels across all categories and exhibits robust performance.

#### 4.5. Effectiveness of abductive methods

In the last part of the experiment, we will select some typical examples to illustrate how our AB-MIML framework operates during the inference process and demonstrate why it outperforms other baseline methods. Detailed information is presented in Fig. 7. We have chosen three representative examples to showcase our method. The first image belongs to the MMP class and is correctly classified by the baseline method, the MIML method, and the AB-MIML method. The second image also falls under the MMP class; however, it is incorrectly classified by the baseline method. The last image represents a healthy periodontal condition, yet both the baseline and MIML methods misclassify it as gingivitis for some reasons. Fig. 7 illustrates the differences in how the baseline method, MIML method, and AB-MIML method arrive at their predictions. Through GRAD-CAM (Zhou et al., 2016; Selvaraju et al., 2017), we visualize how the classification model concentrates on different parts of the input. For the baseline method, the entire periodontal image is treated as a whole, and it utilizes convolution and pooling layers to identify texture patterns in the image that significantly influence the predicted labels. However, in some cases, the model may fail to identify the correct patterns, causing it to focus on the wrong areas of the image and make erroneous judgments. As is shown in the second example, the baseline method focuses on incorrect areas of the input image, resulting in erroneous predictions. In contrast, the MIML method divides the original image into multiple smaller patches. The model identifies patterns to generate feature vectors for each small

region and aggregates these instance vectors using an attention-based approach, ultimately producing the final result through a classifier. This method enables the model to concentrate on local texture features, thereby highlighting key instances and significantly influencing the classification outcome. However, the MIML method still lacks the capability for local error correction; if the model produces an incorrect prediction in a specific region, this can substantially affect the final classification result. The AB-MIML method addresses this limitation by incorporating prior domain knowledge, allowing for some degree of error correction. As demonstrated in the third example, the MIML method aggregates an incorrect classification result due to erroneous local predictions. The abduction block identifies that this classification result conflicts with the knowledge base, as it is unlikely for a patient with gingivitis to have all swelling representations concentrated in the posterior regions of the intraoral image. Therefore, the abduction block calculates the cost of inverting the predicted label based on the confidence of its output and attempts to make adjustments within cost constraints. Ultimately, the algorithm finds that the corrected predictions align better with the constraints of the knowledge base, so that the AB-MIML algorithm outputs the results after flipping the predicted labels. Therefore, the AB-MIML method aligns the prediction results with expert knowledge through the incorporation of domain prior knowledge, thereby enhancing the identification of key instances and improving overall classification performance. This process is consistent with how human doctors make diagnoses.

## 5. Conclusion

In this paper, we propose a novel ABductive Multi-Instance Multi-Label learning framework (AB-MIML) to tackle the complex issue of periodontal disease classification. By leveraging the MIML paradigm, we model the intricate correspondence between images and their corresponding labels, which allows for a many-to-many mapping relationship. Meanwhile, the abduction component integrates expert knowledge expressed by logical formulas into the training and reasoning process, in order to help machine learning make better perceptions. Experimental results demonstrate that our framework achieves SOTA performance in our problem setting, and can be beneficial in numerous practical applications.

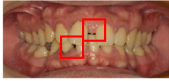
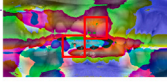
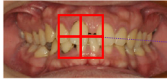
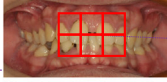
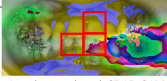
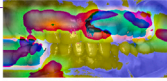
| Raw data                                                                          | Ground truth label and its clues                                                                                     | Inference process under different methods                                                                                  | Comments                                                                                                                                                                     |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <br>label: MMP swelling & recession | <br>Neural computing → label: MMP         | ✓ <ul style="list-style-type: none"><li>Traditional methods</li><li>Manage to concentrate on right regions</li></ul>                                                         |
|                                                                                   |                                                                                                                      | <br>Aggregation of instances → label: MMP | ✓ <ul style="list-style-type: none"><li>MIML methods</li><li>Generate right instance labels</li></ul>                                                                        |
|                                                                                   |                                                                                                                      | <br>Abduce → label: MMP                   | ✓ <ul style="list-style-type: none"><li>ABL-MIML methods</li><li>Instance labels consistent with ground truth</li><li>No abduction changes</li></ul>                         |
|  | <br>label: MMP swelling & recession | <br>Neural computing → label: PH          | ✗ <ul style="list-style-type: none"><li>Traditional methods</li><li>Fail to concentrate on right regions</li></ul>                                                           |
|                                                                                   |                                                                                                                      | <br>Aggregation of instances → label: MMP | ✓ <ul style="list-style-type: none"><li>MIML methods</li><li>Generate right key instance labels</li><li>Some mistake in other instances</li></ul>                            |
|                                                                                   |                                                                                                                      | <br>Abduce → label: MMP                   | ✓ <ul style="list-style-type: none"><li>ABL-MIML methods</li><li>Slightly wrong instance labels</li><li>Correct the label with abductive reasoning</li></ul>                 |
|  | <br>label: PH periodontal health    | <br>Neural computing → label: GI          | ✗ <ul style="list-style-type: none"><li>Traditional methods</li><li>Fail to concentrate on right regions</li><li>Make wrong prediction</li></ul>                             |
|                                                                                   |                                                                                                                      | <br>Aggregation of instances → label: GI  | ✗ <ul style="list-style-type: none"><li>MIML methods</li><li>Generate wrong instance labels</li><li>Make wrong prediction</li></ul>                                          |
|                                                                                   |                                                                                                                      | <br>Abduce → label: PH                  | ✓ <ul style="list-style-type: none"><li>ABL-MIML methods</li><li>Instance labels inconsistent with ground truth</li><li>Correct the label with abductive reasoning</li></ul> |

Fig. 7. Some typical examples to illustrate how our AB-MIML framework works. We select three examples, and each group shows the inference process of the base, MIML and AB-MIML methods respectively. The “comments” part contains additional descriptions on what occurs during the inference process.

CRediT authorship contribution statement

**Zi-Yuan Wu:** Writing – original draft, Methodology, Conceptualization. **Wei Guo:** Writing – original draft, Resources, Data curation. **Wei Zhou:** Resources, Data curation. **Han-Jia Ye:** Writing – review & editing, Investigation, Conceptualization. **Yuan Jiang:** Supervision, Conceptualization. **Houxuan Li:** Supervision, Conceptualization. **Zhi-Hua Zhou:** Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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